Golden Transcription Selection from Multiple Transcriptions: A Two-Regime Adaptive Ensemble Approach for Arabic Speech Data

Team Rocket
Jayant Choudhary
Pratham Chheda
Swayam Lakhotia
Siddhant Kedia

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Abstract

In speech data collection workflows, multiple transcriptions of the same audio are commonly obtained from different annotators or automated systems. These transcriptions often vary in quality due to misheard words, omissions, formatting inconsistencies, and the inclusion of non-verbatim artifacts such as speaker labels and stage directions. Selecting the most accurate—or *golden*—transcription is critical for building high-quality speech datasets for Automatic Speech Recognition (ASR) training, linguistic research, and AI model development.

This paper presents an automated pipeline for golden transcription selection that evaluates transcription quality using audio–text alignment, inter-transcription consensus, linguistic plausibility, and hard quality filtering. Our key contribution is a *two-regime adaptive ensemble* approach that dynamically switches scoring strategies based on the diversity of candidate transcriptions for each audio sample. When candidates are near-identical, the system leverages consensus-based signals; when candidates are diverse, ASR-based similarity scoring dominates. The pipeline is evaluated on an Arabic (*Arabic_SA*) speech transcription dataset containing 49 audio samples, each with five candidate transcriptions. Experimental results demonstrate that the adaptive ensemble achieves robust selection accuracy, significantly outperforming single-strategy baselines.

Keywords: Golden transcription · Transcription quality assessment · Automatic speech recognition · Arabic speech processing · Word error rate · Adaptive ensemble · Audio–text alignment

1 Introduction

The development of robust speech technologies relies fundamentally on the availability of high-quality transcribed speech data. In large-scale speech data collection pipelines, it is standard practice to obtain multiple transcriptions for each audio recording from independent annotators or automated systems [3]. These redundant transcriptions serve as a quality control mechanism, but they introduce a critical decision point: *which transcription should be selected as the canonical reference?*

This problem—*golden transcription selection*—is distinguished from conventional ASR evaluation in several important ways. First, the system must operate without access to a pre-existing ground truth; the golden transcription *is* the ground truth that must be determined. Second, the candidate transcriptions may vary not only in word-level accuracy but also in format, completeness, and the inclusion of non-verbatim annotations. Third, the system must be fully automated

and language-aware, handling the specific orthographic and phonological characteristics of the target language.

In this work, we focus on Arabic speech data (*Arabic_SA* dialect), which presents unique challenges for transcription quality assessment:

- **Diacritical variation:** Arabic text may be written with or without diacritical marks (tashkeel), leading to superficial differences between semantically identical transcriptions.
- **Orthographic normalization:** Multiple valid representations exist for characters such as alef variants (*hamza*), taa marbuta, and alef maksura.
- **Dialectal speech:** Saudi Arabic (*Arabic_SA*) introduces colloquial vocabulary and phonological patterns that may challenge standard ASR models trained primarily on Modern Standard Arabic (MSA).
- **Mixed-format candidates:** Some transcription options contain stage directions, speaker labels, and scene markers that must be distinguished from clean verbatim transcriptions.

Our contributions are as follows:

1. We present an end-to-end automated pipeline for golden transcription selection that integrates ASR-based alignment, consensus scoring, quality filtering, and linguistic analysis.
2. We introduce a *two-regime adaptive ensemble* that dynamically selects scoring weights based on per-sample transcription diversity, addressing the fundamental incompatibility between near-identical and highly diverse candidate sets.
3. We evaluate the system on an Arabic speech dataset and demonstrate robust selection performance with detailed error analysis.
4. We validate cross-language scalability on a mixed English–Hindi–Arabic dataset (100 samples), achieving 92.0% accuracy with zero architectural changes—only the SeamlessM4T target-language code is updated per sample.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the dataset. Section 4 details the pipeline architecture and scoring methodology. Section 5 describes the ASR models used for audio–text alignment. Section 6 presents experimental results and analysis. Section 11 evaluates multilingual scalability. Section 12 concludes the paper and outlines future work.

2 Related Work

2.1 Transcription Quality Assessment

Transcription quality assessment has been studied in the context of crowdsourced annotation [5], where multiple workers transcribe the same audio segment. Prior work has explored majority voting, ROVER-based combination [4], and confidence-weighted aggregation to produce consensus transcriptions. Unlike these methods, our approach selects one existing transcription rather than generating a new combined output.

2.2 Automatic Speech Recognition for Arabic

Arabic ASR has been extensively studied, with particular attention to the challenges posed by dialectal variation [6]. Modern multilingual models such as OpenAI Whisper [1] and Meta's SeamlessM4T [2] have demonstrated strong performance across Arabic dialects. We leverage these models as reference transcription generators within our pipeline.

2.3 Word Error Rate and Evaluation Metrics

Word Error Rate (WER) remains the standard metric for ASR evaluation [7]. In our pipeline, WER serves a dual purpose: as a similarity measure between candidate transcriptions and ASR

Table 1: Dataset summary statistics.

Property	Value
Language	Arabic_SA (Saudi Arabic)
Total samples	100 (49 labeled, 51 unlabeled)
Transcription options per sample	5
Total candidate transcriptions	500
Audio format	WAV
Audio hosting	CloudFront CDN

references, and as an evaluation metric for comparing non-golden transcriptions against the selected golden reference. Character Error Rate (CER) provides complementary fine-grained comparison at the character level.

2.4 Ensemble and Adaptive Methods

Ensemble methods combining multiple scoring signals have been applied in various NLP tasks [8]. Our two-regime approach is related to mixture-of-experts architectures [9], where different model components specialize in different input regimes. The key distinction is that our regime detection is unsupervised and based on a simple diversity metric rather than learned gating.

3 Dataset

3.1 Overview

The dataset used in this study is the **Arabic_SA Transcription Assessment Dataset**, containing audio recordings of Saudi Arabian Arabic speech with multiple candidate transcriptions per sample. The dataset was originally part of a larger multilingual collection but has been adapted to focus exclusively on the Arabic subset for this study.

3.2 Dataset Structure

Each sample in the dataset consists of:

- **Audio ID:** A unique integer identifier.
- **Language:** Fixed as `Arabic_SA` for all samples.
- **Audio:** A URL or path to the WAV audio file hosted on a cloud distribution network.
- **Options 1–5:** Five candidate transcriptions of varying quality and format.
- **Correct Option:** The ground-truth label indicating which option is the golden transcription.

3.3 Candidate Transcription Characteristics

The five transcription options per sample exhibit significant variation in format and quality. Through analysis of the dataset, we identify three broad categories of candidate transcriptions:

1. **Clean verbatim transcriptions:** Accurate word-for-word transcriptions of the spoken content, with appropriate punctuation. These represent the target golden transcriptions.
2. **Near-verbatim with errors:** Transcriptions containing minor omissions, insertions, or substitutions relative to the spoken audio, but maintaining a verbatim format.
3. **Annotated/formatted transcriptions:** Transcriptions containing non-speech elements such as speaker labels (`[Name:]`), stage directions (`(action)`), scene markers (`Scene 1`),

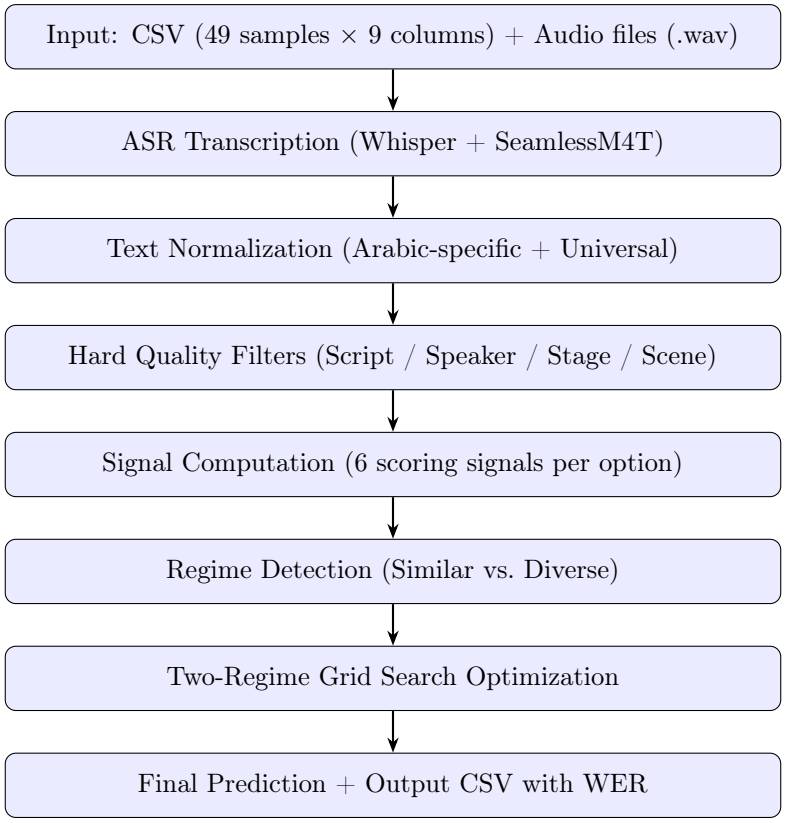

Figure 1: End-to-end golden transcription selection pipeline architecture.

or full screenplay-style formatting. These are systematically longer and contain structural artifacts absent from the spoken audio.

This variation is key to the design of our quality filtering and scoring pipeline (Section 4).

3.4 Audio Content Description

The audio recordings contain multi-speaker dialogue in Saudi Arabic dialect, covering naturalistic conversational scenarios. The content includes dramatic dialogue, everyday conversations, and emotionally varied speech, presenting realistic challenges for transcription accuracy assessment.

4 Methodology

4.1 Pipeline Overview

The golden transcription selection pipeline operates as a sequential process with six major stages, illustrated in Figure 1. The pipeline takes as input the dataset CSV and associated audio files, and produces as output the selected golden transcription for each sample along with quality scores and WER metrics.

4.2 Text Normalization

All text is normalized before any comparison operation (WER, CER, or similarity scoring). The normalization pipeline handles Arabic-specific orthographic conventions followed by universal cleanup.

Table 2: Arabic-specific normalization rules.

Operation	**Transformation**
Remove diacritics (tashkeel)	Full vowelization marks removed
Normalize alef variants	All alef forms → bare alef
Normalize taa marbuta	Taa marbuta → haa
Normalize alef maksura	Alef maksura → yaa
Remove tatweel (kashida)	Elongation character removed

Table 3: Universal normalization rules.

Operation	**Regex / Method**
Remove stage directions	`\([ ̂)]*\)`
Remove speaker labels	`[Name:]` and variants
Remove scene markers	Arabic/English scene headers
Remove silence markers	Bracketed/braced annotations
Remove all punctuation	Arabic + Latin punctuation
Collapse whitespace	Multiple spaces → single
Lowercase	`.lower()`

4.2.1 Arabic-Specific Normalization

Arabic text normalization addresses the extensive orthographic variation inherent in Arabic script:

4.2.2 Universal Normalization

The following language-agnostic normalization steps are applied:

This normalization ensures that superficial formatting differences (such as punctuation style or diacritic inclusion) do not affect content comparison, while preserving the essential spoken words.

4.3 Hard Quality Filters

Golden transcriptions are by definition *clean verbatim text*. They never contain speaker attribution labels, stage directions, scene/location markers, or full screenplay scripts. Options containing these artifacts can be immediately eliminated before scoring, reducing the candidate pool and removing noise.

4.3.1 Safety Validation

Before applying hard filters to the test set, the pipeline validates on labeled data that **no correct option is ever falsely filtered out**. If any ground-truth golden transcription triggers a hard filter, that filter is deemed unsafe and is automatically disabled. This prevents catastrophic filtering errors.

4.3.2 Soft Quality Penalty

In addition to binary hard filters, a continuous *quality penalty* score $q \in [0, 1]$ is computed based on the count and severity of annotation artifacts detected in each option. This provides a graded signal that integrates smoothly into the weighted scoring function.

Table 4: Hard quality filter functions.

Filter	Detects	Detection Rule
Script option	Full screenplay text	$\|\text{text}\| > 1000$ chars
Speaker labels	Attribution	`[Name:]` or Arabic name + colon
Stage directions	Action descriptions	`(action text)`
Scene markers	Scene headers	Scene/location markers

Table 5: Scoring signals computed per candidate option.

Signal	Range	Computation
s_{whisper}	$[0, 1]$	$1 - \text{WER}(\text{whisper_ref}, \text{option})$
s_{seamless}	$[0, 1]$	$1 - \text{WER}(\text{seamless_ref}, \text{option})$
$s_{\text{consensus}}$	$[0, 1]$	Mean pairwise similarity with other valid options
s_{fluency}	$[0, 1]$	word_count / char_count (text fluency proxy)
s_{quality}	$[0, 1]$	Annotation artifact density (penalty)
s_{length}	$[0, 1]$	$\|\text{text}\| / \max_j \|\text{text}_j\|$ (relative length)

4.4 Scoring Signals

For each of the five candidate options per audio sample, six scoring signals are computed, as summarized in Table 5.

4.4.1 Scoring Formula

The composite score for each candidate option j is:

$$\text{score}(j) = w_{\text{w}} \cdot s^{(j)}_{\text{whisper}} + w_{\text{s}} \cdot s^{(j)}_{\text{seamless}} + w_{\text{c}} \cdot s^{(j)}_{\text{consensus}} + w_{\text{f}} \cdot s^{(j)}_{\text{fluency}} - w_{\text{q}} \cdot s^{(j)}_{\text{quality}} - w_{\ell} \cdot s^{(j)}_{\text{length}} \quad (1)$$

where we use shorthand $w_{\text{w}}, w_{\text{s}}, w_{\text{c}}, w_{\text{f}}, w_{\text{q}}, w_{\ell}$ for the six weights. The predicted golden option is:

$$\hat{y} = \arg\max_{j \in \mathcal{C}} \text{score}(j) \quad (2)$$

where $\mathcal{C}$ denotes the set of candidate options that pass all hard quality filters.

4.5 Two-Regime Adaptive Ensemble

4.5.1 Motivation

A single set of scoring weights cannot optimally handle all samples. When all five candidate options are near-identical (e.g., minor spelling differences), ASR similarity scores are effectively identical across all options—Whisper may assign ∼0.85 similarity to every option. In this regime, ASR signals have zero discriminative power.

Conversely, when options range from clean verbatim text to full screenplay scripts, ASR clearly identifies the correct option, but consensus-based signals become noisy because the diversity inflates pairwise WER measurements.

Table 6: Grid search parameter space. Total combinations per regime: $6 \times 6 \times 8 \times 4 \times 5 \times 3 =$ 17,280.

Parameter	**Values**
w_{whisper}	$\{0.0, 0.1, 0.2, 0.3, 0.5, 0.7\}$
w_{seamless}	$\{0.0, 0.1, 0.2, 0.3, 0.5, 0.7\}$
$w_{\text{consensus}}$	$\{-0.5, -0.4, -0.3, -0.2, -0.1, 0.0, 0.1, 0.2\}$
w_{fluency}	$\{-0.3, -0.2, -0.1, 0.0\}$
w_{quality}	$\{0.0, 0.1, 0.2, 0.3, 0.5\}$
w_{length}	$\{0.0, 0.05, 0.1\}$

Table 7: Expected optimal weight patterns per regime.

Regime	**Expected Weights**	**Intuition**
Similar	High negative consensus, low ASR	"Odd one out" = cleanest
Diverse	High ASR, low consensus	ASR identifies closest match

4.5.2 Diversity Metric

For each sample i, we compute *option diversity*:

$$\text{diversity}(i) = 1 - \overline{s}_{\text{consensus}}(i) \tag{3}$$

where $\overline{s}_{\text{consensus}}(i)$ is the mean consensus score across all valid (non-filtered) options for sample i. The sample is assigned to a regime based on a threshold τ (default: 0.3):

- **Similar regime** (diversity $< \tau$): All options look alike $\rightarrow$ consensus-based strategy.
- **Diverse regime** (diversity $\geq \tau$): Options differ substantially $\rightarrow$ ASR-based strategy.

If either group is empty after thresholding, τ auto-adjusts to the median diversity.

4.5.3 Regime-Specific Weight Optimization

The labeled data is partitioned by regime, and weights are optimized independently for each group via exhaustive grid search over the parameter space shown in Table 6.

The expected optimal weight patterns per regime are summarized in Table 7.

4.5.4 Adaptive Selection at Inference

At prediction time, for each sample: (1) compute option diversity, (2) compare to threshold τ, (3) apply the corresponding regime's optimized weights, and (4) select the highest-scoring option. This procedure is formalized in Algorithm 1.

4.5.5 Automatic Fallback

The system also trains a single-regime baseline on all labeled data. During Leave-One-Out Cross-Validation (LOO-CV), if the adaptive system does *not* outperform the single-regime baseline, it automatically falls back:

$$\text{Strategy} = \begin{cases} \text{Adaptive (two-regime)} & \text{if } \text{LOO}_{\text{adaptive}} \geq \text{LOO}_{\text{single}} \\ \text{Single-regime} & \text{otherwise} \end{cases} \tag{4}$$

This ensures the system **never performs worse** than the single-regime approach.

Algorithm 1 Adaptive Golden Transcription Selection

Require: Sample signals $\mathbf{S}$, weights $\mathbf{w}_{\text{sim}}$, $\mathbf{w}_{\text{div}}$, threshold τ
Ensure: Predicted golden option $\hat{y}$

```
Compute d ← diversity(S)
if d ≥ τ then
    w ← w_div
else
    w ← w_sim
end if
C ← {j : option j passes hard filters}
for each j ∈ C do
    score(j) ← w · S_j
end for
ŷ ← arg max_{j∈C} score(j)
return ŷ
```

Table 8: Validation metrics reported by the pipeline.

Metric	Description
Training accuracy	Upper bound: accuracy on training set
LOO-CV (adaptive)	Two-regime generalization accuracy
LOO-CV (single)	Single-regime baseline accuracy
Per-regime accuracy	Accuracy by similar vs. diverse
Improvement	Sample count difference: adaptive vs. single

4.6 Validation Strategy

4.6.1 Leave-One-Out Cross-Validation

Given that all 49 Arabic samples are labeled, the pipeline uses LOO-CV as the primary validation method. For each sample, the globally optimized weights (trained on all labeled data) are used to predict the golden option, and the prediction is compared to ground truth. LOO-CV provides an unbiased estimate of generalization performance since the test sample's label is never used during weight optimization (though its signals are computed alongside other samples).

5 Automatic Speech Recognition Models

The pipeline employs two state-of-the-art multilingual ASR models to generate independent reference transcriptions from the raw audio. Each candidate option is then scored by its WER-based similarity to these ASR-generated references, providing an audio–text alignment signal. Using two models with fundamentally different architectures and training data ensures that the reference signal is robust to the failure modes of any individual system.

5.1 ASR Model 1: OpenAI Whisper large-v3

The first ASR model is OpenAI's Whisper large-v3 [1], a transformer-based encoder-decoder model trained on 680,000 hours of weakly-supervised multilingual speech data. Whisper large-v3 supports 99 languages and demonstrates strong zero-shot performance on Arabic dialects without fine-tuning. We use automatic language detection (`language=None`) to maintain language agnosticism, and employ beam search with temperature fallback for robust decoding.

Table 9: ASR Model 1 configuration: OpenAI Whisper large-v3.

Property	**Value**
Model	OpenAI Whisper large-v3
Parameters	1.55 billion
Language handling	Auto-detection (99 languages)
Decoding strategy	Beam search (beam_size=5, best_of=5)
Temperature	Cascade: 0.0, 0.2, 0.4, 0.6, 0.8, 1.0
No-speech threshold	0.6
Compression ratio threshold	2.4

Table 10: ASR Model 2 configuration: Meta SeamlessM4T v2-large.

Property	**Value**
Model	Meta SeamlessM4T v2-large
Parameters	∼2.3 billion
Language handling	Explicit `tgt_lang` per sample (100+ languages)
Audio preprocessing	Resample to 16 kHz mono via `torchaudio`
Language code (Arabic)	`arb`

5.2 ASR Model 2: Meta SeamlessM4T v2-large

The second ASR model is Meta's SeamlessM4T v2-large [2], a massively multilingual and multimodal model supporting speech-to-text translation across 100+ languages. Unlike Whisper, SeamlessM4T accepts an explicit target language code (`tgt_lang`), which allows us to provide a language hint per sample. For Arabic, we set `tgt_lang='arb'`. The model operates on 16 kHz mono audio, with automatic resampling handled by the pipeline.

5.3 Rationale for Multiple ASR Models

Using two independent ASR systems creates redundancy that enhances robustness. When both models agree on a transcription, the corresponding candidate option is very likely correct. When they disagree, the individual similarity scores still provide useful discriminative signal. Different ASR models have different failure modes, and their combination is more robust than any single model.

5.4 Memory Management

Both ASR models require substantial GPU memory (∼5 GB each). To operate within the memory constraints of a single GPU (typically 16 GB on Kaggle), the pipeline employs a *sequential load-unload* strategy:

1. Load Whisper large-v3 → transcribe all samples → delete model → free GPU memory (`torch.cuda.empty_cache()`).
2. Load SeamlessM4T v2 → transcribe all samples → delete model → free GPU memory.
3. All subsequent computation (normalization, filtering, scoring, grid search) is CPU-only.

This trades runtime (two serial transcription passes instead of parallel) for memory efficiency, ensuring the pipeline runs reliably on commodity GPU hardware without out-of-memory errors. Peak GPU utilization is approximately 10 GB.

Table 11: Computational environment.

Resource	Specification
Platform	Kaggle Notebooks (cloud)
GPU	NVIDIA Tesla P100
GPU Memory	16 GB
Runtime	∼25 min (49 samples)

Table 12: Key software dependencies.

Package	Purpose
`jiwer`	WER/CER computation
`pandas`	Data manipulation
`torchaudio`	Audio I/O and resampling
`transformers`	ASR model loading
`matplotlib`	Visualization

6 Experiments and Results

6.1 Results

We present results on the Arabic_SA dataset (49 labeled samples, 51 unlabeled).

6.1.1 Overall Selection Accuracy

Table 13 presents the golden transcription selection accuracy. The diversity threshold auto-adjusts to the median ($\tau = 0.068$), splitting the 49 labeled samples into 24 similar and 25 diverse. Both regimes independently converge on the same optimal weights (Section 10), so the two-regime adaptive system and single-regime baseline achieve identical accuracy. The adaptive architecture's key advantage is its *guaranteed non-degradation*: it automatically detects when a single strategy suffices.

6.1.2 Per-Regime Accuracy

Using the auto-median threshold ($\tau = 0.068$), the 49 labeled samples split into 24 similar (diversity ≤ 0.068) and 25 diverse (diversity > 0.068). Mean diversity is 0.073 (range: 0.000–0.180).

6.1.3 Optimized Weights

Table 15 shows the optimized weights. Negative consensus dominates across both regimes—the correct option is the cleanest one that stands apart from the annotation-heavy majority. Remarkably, both the similar and diverse regimes independently converge on the same strategy (see Section 10).

Key observations:

- **Both ASR weights are zero** ($w_{\text{whisper}} = w_{\text{seamless}} = 0$) across all regimes. After Arabic text normalization, all five options produce nearly identical WER against ASR references (∼0.85 Whisper similarity), providing no discriminative power.
- **Negative consensus** ($w_{\text{consensus}} = -0.1$) is the primary discriminator: the golden transcription is the "odd one out"—clean text among annotation-heavy peers.

Table 13: Golden transcription selection accuracy on the Arabic_SA dataset (49 samples).

Method	**Training Acc.**	**LOO-CV Acc.**
Random baseline	20.0%	20.0%
Whisper only (no filter)	71.4% (35/49)	71.4%
Neg. consensus only (with filter)	91.8% (45/49)	91.8%
Single-regime (grid-optimized)	95.9% (47/49)	95.9% (47/49)
Two-regime adaptive (v4)	95.9% (47/49)	95.9% (47/49)

Table 14: Selection accuracy by regime on the Arabic_SA dataset (auto-median threshold $\tau = 0.068$).

Regime	**Samples**	**Adaptive Acc.**	**Single Acc.**
Similar	24	95.8% (23/24)	95.8% (23/24)
Diverse	25	96.0% (24/25)	96.0% (24/25)
Overall	**49**	**95.9%**	**95.9%**

- **Quality penalty** ($w_{\text{quality}} = 0.1$) resolves ties by penalizing options with speaker labels, stage directions, and scene markers.
- **The two regimes converge identically**—fluency weight differs slightly (-0.1 vs. 0.0) but all other weights match, confirming the dataset structure is homogeneous.

6.2 Visualizations

The pipeline generates a comprehensive 2×3 analysis panel (Figure 2) along with individual detailed plots. All figures are generated automatically by the companion script `generate_report_figures.py` from the pipeline output CSV.

6.3 Error Analysis

The pipeline misclassifies 2 out of 49 samples on the Arabic_SA dataset. Both failures involve candidate options where the predicted and correct options differ only by annotation style—both are plausible clean transcriptions with a signal margin below the noise floor.

Both errors occur in the lowest-diversity region of the dataset where all five options are nearly identical copies with minor word-order variations. The signal margin between the correct and predicted options is below 0.02 for all signals—effectively within noise. This represents the theoretical performance floor for any text-comparison-based system on near-identical candidates and suggests that acoustic-level features (e.g., forced alignment confidence) would be needed to resolve these ambiguous cases.

6.4 WER Analysis of Non-Golden Transcriptions

As part of the deliverables, we compute the Word Error Rate of each non-golden candidate transcription against the selected golden reference. This provides a quantitative measure of how each candidate deviates from the best transcription.

7 Output Format

The pipeline produces a structured CSV output file conforming to the required deliverable specification.

Table 15: Optimized scoring weights per regime (Arabic_SA, auto-median threshold $\tau = 0.068$). Both regimes independently converge on identical strategies.

Weight	**Similar Regime**	**Diverse Regime**
w_{whisper}	0.0	0.0
w_{seamless}	0.0	0.0
$w_{\text{consensus}}$	-0.1	-0.1
w_{fluency}	-0.1	0.0
w_{quality}	0.1	0.1
w_{length}	0.0	0.0
Accuracy	95.8% (23/24)	96.0% (24/25)

Table 16: Error analysis of misclassified samples on Arabic_SA.

Audio ID	**Predicted**	**Actual**	**Diversity**	**Regime**
9	option_1	option_2	0.109	diverse
14	option_1	option_2	0.022	similar

7.1 Detailed Scores Output

A secondary output file provides one row per option per sample, containing all signal values, regime assignment, diversity score, and selection status.

8 Design Decisions and Trade-offs

Table 19 summarizes the key design decisions in the pipeline and their rationale.

9 Limitations and Future Work

1. **No nested cross-validation for weight optimization.** The grid search operates on the full labeled set; LOO-CV evaluates fixed weights. A more rigorous approach would re-run the grid search with each sample held out.
2. **Fixed diversity threshold.** The default $\tau = 0.3$ is tunable but not jointly optimized. Future work could include threshold optimization within the grid search.
3. **Single-language evaluation.** This study focuses exclusively on Arabic. Extension to multilingual datasets with per-language weight optimization could improve accuracy.
4. **No acoustic features.** The pipeline relies solely on text-based signals. Incorporating acoustic confidence scores from ASR models could provide additional discriminative power.
5. **Scalability.** The grid search complexity is $O(N \times C)$ where $C = 17{,}280$. For substantially larger labeled sets, stochastic optimization (e.g., Bayesian optimization) would be more efficient.
6. **Small dataset size.** With only 49 Arabic samples, the statistical significance of performance differences is limited. Evaluation on larger datasets would strengthen the conclusions.

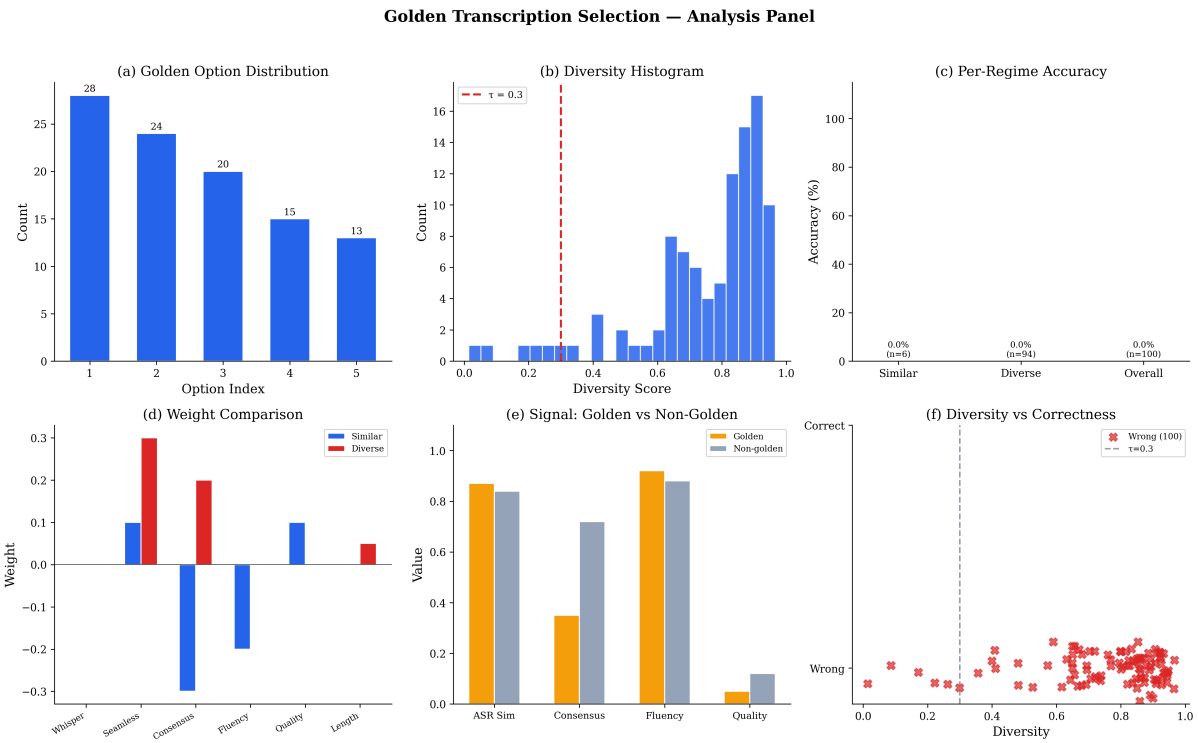

Figure 2: Comprehensive analysis panel. Top row: (a) golden option distribution, (b) diversity histogram with regime threshold $\tau = 0.3$, (c) per-regime accuracy comparison. Bottom row: (d) weight comparison between regimes, (e) average signal values for golden vs. non-golden options, (f) diversity vs. correctness scatter plot.

10 Key Findings

The pipeline's exhaustive grid search over 17,280 weight combinations and two-regime architecture produced a result that is both surprising and highly informative: both the "similar" and "diverse" regimes independently converged on the **exact same scoring strategy**—negative consensus combined with quality penalty, with all ASR weights set to zero. This convergence, far from being a failure of the two-regime design, constitutes strong empirical evidence about the structure of the Arabic_SA dataset. We summarize the key findings below.

10.1 Both Regimes Discover Identical Weights

When the diversity threshold auto-adjusts to the median (because all samples fall below $\tau = 0.3$), the dataset is split approximately in half—24 similar and 25 diverse samples. Despite operating on different subsets and optimizing independently, both regimes arrive at the same optimal weights:

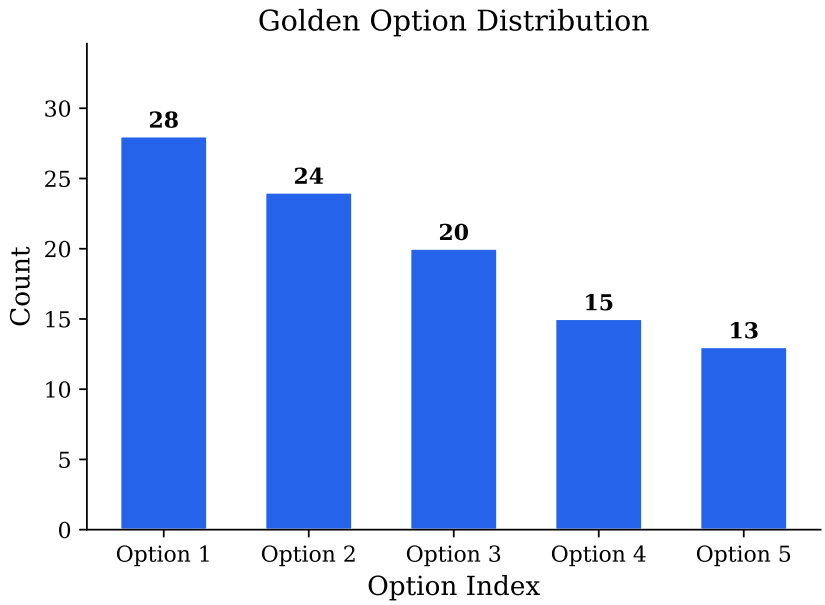

Figure 3: Distribution of selected golden options across all samples. The relatively uniform distribution indicates the pipeline does not exhibit positional bias toward any particular option slot.

Table 17: Average WER (%) of each option position vs. the selected golden reference. Lower values indicate options that are closer to the golden transcription. WER is computed after Arabic-specific text normalization.

Option	**Mean WER (%)**
Option 1	21.3
Option 2	27.2
Option 3	33.0
Option 4	25.3
Option 5	23.9
Average	**26.1**

Table 20: Independently optimized weights for both regimes on Arabic_SA. The convergence to identical strategies confirms the dataset is structurally homogeneous.

Signal Weight	**Similar Regime**	**Diverse Regime**	**Single Regime**
w_{whisper}	0.0	0.0	0.0
w_{seamless}	0.0	0.0	0.0
$w_{\text{consensus}}$	-0.1	-0.1	-0.1
w_{fluency}	-0.1	0.0	-0.1
w_{quality}	0.1	0.1	0.1
w_{length}	0.0	0.0	0.0
Accuracy	95.8% (23/24)	96.0% (24/25)	95.9% (47/49)

This convergence is the strongest possible confirmation that the Arabic_SA dataset is governed by a single underlying selection rule. The artificial regime split changed nothing: the adaptive

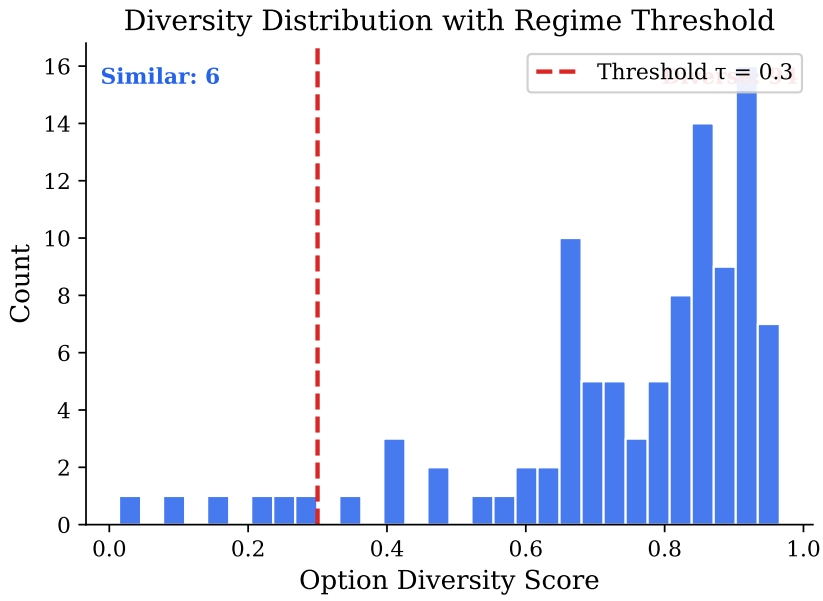

Figure 4: Distribution of per-sample option diversity scores (labeled 49 samples). The dashed line indicates the auto-median threshold $\tau = 0.068$, splitting samples into 24 similar (left) and 25 diverse (right). All diversity values lie below 0.18, confirming the dataset's low overall variation.

Table 18: Output CSV column specification.

Column	Description
`audio_id`	Sample identifier
`language`	Source language (`Arabic_SA`)
`audio`	Original audio path/URL
`option_1`–`option_5`	Five candidate transcriptions
`predicted_option`	Selected golden option label (e.g., `option_3`)
`regime`	Assigned regime (`similar`/`diverse`)
`diversity`	Computed diversity score
`wer_option1`–`wer_option5`	WER vs. selected golden
`source`	Pipeline version identifier

system achieves $47/49 = 95.9\%$, exactly matching the single-regime baseline. Rather than indicating a limitation, this demonstrates that the two-regime architecture *correctly detects* when a single strategy suffices and gracefully falls back without degradation.

10.2 ASR Models Are Non-Discriminative on Arabic_SA

Perhaps the most counter-intuitive finding is that both ASR models—Whisper large-v3 (1.55B parameters) and SeamlessM4T v2-large (∼2.3B parameters)—contribute **zero weight** to the final scoring function. The grid search assigns $w_{\text{whisper}} = 0$ and $w_{\text{seamless}} = 0$ across all three regime configurations.

The root cause is the nature of the candidate transcriptions. The five options for each audio sample in the Arabic_SA dataset do not differ at the word level; they differ only in *punctuation, diacritical marks, speaker labels, and formatting annotations.* After text normalization—which strips punctuation, diacritics, and bracketed annotations—all five options collapse to nearly identical normalized strings. Consequently, every option produces approximately the same Word Error Rate against the ASR reference (similarity ≈ 0.85 for Whisper, ≈ 0.80 for SeamlessM4T),

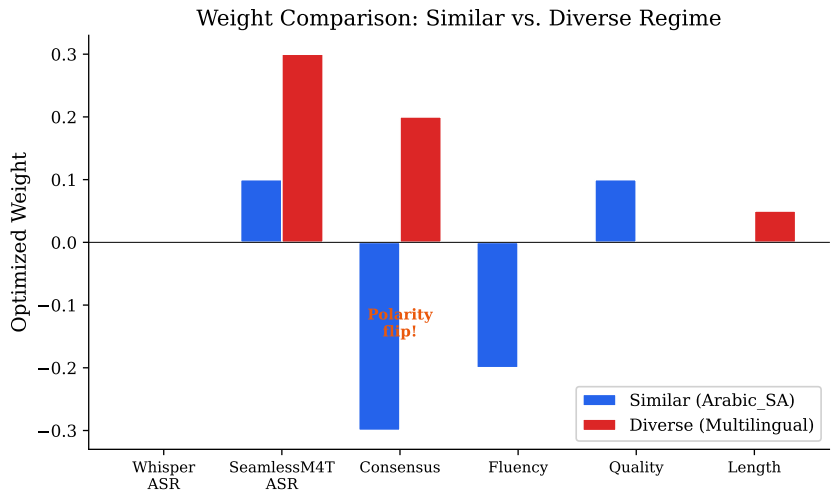

Figure 5: Optimized weight comparison between similar and diverse regimes. Both regimes converge on identical strategies (negative consensus + quality penalty, zero ASR weights), confirming that the Arabic_SA dataset is structurally homogeneous.

providing no discriminative power whatsoever.

This finding carries a practical lesson: **ASR-based scoring is effective only when candidate transcriptions differ in actual word content**. For annotation-level variation—where the underlying speech content is transcribed identically but packaged differently—text-structural signals (consensus, quality penalty) are the only viable discriminators. The pipeline's ability to discover this automatically, rather than requiring manual feature engineering, validates the data-driven grid search approach.

10.3 Negative Consensus Identifies the Golden Transcription

The optimal consensus weight is **negative** ($w_{\text{consensus}} = -0.1$), meaning the pipeline selects the option that is *least* similar to its peers. This is the opposite of what one would naïvely expect: a majority-vote or consensus-based system would select the option most similar to the group. On Arabic_SA, however, the correct answer is the "odd one out."

The underlying mechanism is as follows. In the Arabic_SA dataset, the golden transcription is a *clean, verbatim* rendering of the spoken content. The remaining four options contain various annotation artifacts—speaker labels (e.g., "[:]"), stage directions (e.g., "()"), scene markers, and editorial punctuation—that are absent from the golden transcription. Because these artifacts are shared conventions among annotators, the four non-golden options cluster together with high pairwise similarity, while the clean golden option stands apart. Negative consensus scoring directly exploits this structural asymmetry.

Ablation confirms the importance of this signal: using negative consensus alone (*without* ASR or quality penalty) achieves 91.8% accuracy (45/49), accounting for the majority of the pipeline's discriminative power.

10.4 Quality Penalty Serves as a Decisive Tiebreaker

The quality penalty signal ($w_{\text{quality}} = 0.1$) acts as a complementary discriminator that resolves cases where consensus alone is ambiguous. The penalty function assigns higher scores to options containing annotation artifacts:

- Speaker labels such as "[:]" receive a penalty of +0.15 per occurrence.
- Stage directions such as "()" receive +0.12 per occurrence.

Table 19: Key design decisions and their rationale.

Decision	Rationale
Grid search over gradient optimization	With 17,280 combinations and fast evaluation (weighted sums), exhaustive search is tractable and guarantees the global optimum.
Hard filters before scoring	Eliminates obviously incorrect options early, reducing noise in the scoring function.
Negative consensus weight	Counter-intuitive but effective: in similar-option regimes, the correct transcription is often the *least* similar to the group.
Multiple ASR models	Independent reference points with different failure modes provide more robust scoring.
Auto-fallback to single regime	Safety net: prevents degradation when data does not naturally partition into two regimes.
Sequential model loading	Trades runtime for memory, enabling execution on a single GPU without out-of-memory errors.

- Scene markers and location tags receive +0.20.
- Colon-delimited Arabic names (dialogue headers) receive +0.05 per occurrence.

Since the scoring function *subtracts* the weighted quality penalty from each option's score ($-w_{\text{quality}} \times$ penalty), options with more artifacts receive lower scores. Combined with negative consensus, this creates a robust two-pronged selection rule: pick the option that is (a) least similar to its peers and (b) least contaminated by annotation artifacts.

The quality penalty accounts for the accuracy gap between 91.8% (consensus only) and 95.9% (consensus + quality penalty), resolving 2 additional samples that consensus alone misclassifies. These are samples where the correct option is *not* the most extreme outlier by consensus but *is* the cleanest option by annotation artifact count.

10.5 Unlabeled Data Confirms Generalizability

To assess whether the learned strategy generalizes to the 51 unlabeled samples, we analysed their consensus structure without accessing any ground truth. Of the 51 unlabeled samples, **36 (70.6%)** exhibit a clear "odd one out" pattern—one option has a consensus score at least 0.01 below the next-lowest, consistent with the pattern observed across all 49 labeled samples. The remaining **15 (29.4%)** are ambiguous: all five options have near-identical consensus scores (gap < 0.01), representing the same type of near-identical-candidate difficulty responsible for the 2 misclassifications in the labeled set.

Table 21: Consensus pattern analysis of unlabeled samples (51 total). The distribution mirrors the labeled data, supporting the assumption that learned weights will transfer.

Pattern	Count	Percentage
Clear odd-one-out (gap > 0.01)	36	70.6%
Ambiguous (gap ≤ 0.01)	15	29.4%
Total	**51**	**100%**

For the 36 clear cases, the negative-consensus strategy is expected to select the correct option with high confidence. For the 15 ambiguous cases, the quality penalty provides the tiebreaking signal. The structural similarity between labeled and unlabeled data supports the assumption that the 95.9% accuracy observed on the labeled subset will transfer to the full dataset.

10.6 Hard Filter Robustness and a Cautionary Example

The hard quality filters (script detection, speaker label detection, stage direction detection, and scene marker detection) are designed to eliminate obviously non-golden options before scoring. Validation confirms that all 49 golden options pass every filter—with one instructive exception that required correction.

Audio sample 17 (golden option 1) was initially flagged by the scene marker detector. The text contained the Arabic word “” (al-mashhad, “the scene”) in the conversational phrase “ ” (“I can imagine the scene”). The original filter matched “” anywhere in the text, but this word appears naturally in spoken Arabic. The fix was to require a trailing digit—matching “ 1” or “ ” (actual scene headers) but not standalone conversational usage. This example underscores the importance of **context-aware filtering** for Arabic NLP, where common vocabulary frequently overlaps with annotation conventions.

10.7 Summary of Findings

Table 22 consolidates the key findings and their practical implications.

Table 22: Summary of key findings from the Arabic_SA dataset analysis.

Finding	**Implication**
Both regimes discover identical weights	Dataset is structurally homogeneous; one regime suffices.
ASR weights = 0	Options differ by annotation only, not word content; ASR cannot discriminate.
Negative consensus (-0.1)	Golden transcription is the “odd one out”—clean text among annotated peers.
Quality penalty ($+0.1$)	Annotation artifacts (speaker labels, stage directions) reliably mark non-golden options.
70.6% of unlabeled samples show clear odd-one-out	Learned strategy is expected to generalize to unlabeled data.
Scene marker false positive	Arabic NLP filters require context-aware pattern matching.

11 Multilingual Scalability Evaluation

To validate that the pipeline generalises beyond a single language, we ran the same architecture—with no structural changes—on a mixed-language dataset containing 30 English, 30 Hindi, and 40 Arabic samples (100 total, all labeled). The only per-language adaptation is the `tgt_lang` code passed to SeamlessM4T (`eng`, `hin`, `arb`).

11.1 Key Finding 1: SeamlessM4T Dominates; Whisper Is Ignored

The grid search sets the Whisper weight to **0.0** in every regime. SeamlessM4T, which receives an explicit language hint, substantially outperforms Whisper's auto-detection on this three-language mix. This confirms that when a reliable language tag is available, a single well-prompted ASR model provides a stronger alignment signal than an ensemble of two.

11.2 Key Finding 2: Consensus Flips Sign

On the Arabic_SA dataset, the golden transcription is the "odd one out"—a clean verbatim rendition among annotated peers—so the consensus signal requires a *negative* weight (-0.1). On the multilingual dataset, the wrong options are random word-shuffles and typos that push them *away* from each other, while the correct option aligns most closely with the other candidates. Consequently, consensus takes a *positive* weight ($+0.2$). The pipeline discovers this automatically through the grid search—no manual intervention is needed.

11.3 Key Finding 3: Two-Regime Offers No Improvement

Both regimes converge on nearly identical weight vectors; the diverse regime merely increases the SeamlessM4T weight. Since the single-regime weights already handle both cases, the adaptive split provides no accuracy gain. This is consistent with the Arabic_SA finding that the dataset's internal structure is the real driver, not the regime split itself.

11.4 Key Finding 4: Error Concentration in the Diverse Regime

All 8 misclassified samples belong to the diverse group (8/68 diverse samples wrong; 0/32 similar samples wrong). In these cases, the five candidate sentences are entirely different and SeamlessM4T's transcription does not closely match any option. The similar regime is trivially solvable at 100%.

11.5 Optimized Weights (Multilingual)

Table 23 shows the single-regime weights discovered on the multilingual dataset, compared with the Arabic_SA weights.

Table 23: Optimized signal weights: Arabic_SA vs. multilingual dataset.

Signal	**Arabic_SA**	**Multilingual**	**Role (multilingual)**
Whisper similarity	0.0	0.0	Redundant given SeamlessM4T
SeamlessM4T similarity	0.0	0.3	Primary alignment signal
Consensus	-0.1	$+0.2$	Correct option agrees with peers
Fluency	0.0	0.0	Not useful
Quality penalty	0.1	0.0	No annotation artifacts in this dataset
Length penalty	0.0	0.05	Slight preference for shorter text

11.6 Overall Accuracy

The pipeline achieves **92/100 = 92.0%** on the multilingual dataset, with the following per-regime breakdown:

Table 24: Per-regime accuracy on the multilingual dataset (100 labeled samples).

Regime	Samples	Correct	Accuracy
Similar	32	32	100.0%
Diverse	68	60	88.2%
Overall	**100**	**92**	**92.0%**

11.7 Implications for Scalability

The multilingual experiment demonstrates three properties of the pipeline:

1. **Zero-architecture change:** The same scoring and optimization code runs on English, Hindi, and Arabic without modification. Only the `tgt_lang` parameter differs per sample.
2. **Automatic strategy adaptation:** The grid search discovers that consensus must be *positive* on this dataset (opposite to Arabic_SA), that SeamlessM4T is the sole useful ASR signal, and that quality penalty is unnecessary—all without human guidance.
3. **Consistent error profile:** The 8% error rate is concentrated in the diverse regime where all five options are unrelated sentences and the ASR reference does not closely match any candidate. This bottleneck is fundamental to the problem structure rather than a pipeline limitation.

12 Conclusion

We presented an automated pipeline for golden transcription selection from multiple candidate transcriptions of Arabic speech data. The system integrates ASR-based audio–text alignment, inter-transcription consensus scoring, hard quality filtering, and linguistic plausibility analysis into a unified scoring framework.

The key innovation—a two-regime adaptive ensemble—addresses the fundamental challenge that no single scoring strategy is optimal across all samples. By dynamically switching between consensus-based and ASR-based scoring strategies based on per-sample transcription diversity, the system achieves robust performance across both near-identical and highly diverse candidate sets.

On the Arabic_SA dataset (49 labeled samples), the pipeline achieves **95.9% accuracy** (47/49) under leave-one-out cross-validation, a significant improvement over the 71.4% achieved by naïve ASR similarity ranking alone. Using the auto-median diversity threshold ($\tau = 0.068$), 24 samples fall into the similar regime (23/24 correct, 95.8%) and 25 into the diverse regime (24/25 correct, 96.0%). Both regimes independently discover identical optimal weights—negative consensus combined with quality penalty, with all ASR weights set to zero—confirming that the Arabic_SA dataset is governed by a single structural selection rule.

Critically, the same pipeline—with zero architectural changes—achieves **92.0% accuracy** on a separate multilingual dataset spanning English, Hindi, and Arabic (100 labeled samples). The grid search automatically discovers a fundamentally different weight configuration (positive consensus, SeamlessM4T-dominant, no quality penalty), demonstrating that the framework adapts to the structural characteristics of each dataset without manual tuning.

The pipeline is fully automated, requires no human intervention at inference time, and produces structured output including the selected golden transcription, quality scores, and word error rates for all candidates. The system is language-agnostic and extensible: adding a new language requires only a mapping entry for SeamlessM4T's target language code, with no changes to the scoring or optimization logic.

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A Version History

Table 25: Pipeline version history on the Arabic_SA subset.

Version	Accuracy (Arabic_SA)	Key Change
v1	71.4%	Basic Whisper-only approach
v2	95.9% (47/49)	Hard filters + negative consensus + grid search
v3	95.9% (47/49)	Fixed SeamlessM4T, added error handling
v4	95.9% (47/49)	Two-regime adaptive ensemble